Q-OPS: Evidence-Aware Quantum–Classical Routing for Trustworthy Human–AI Staffing

Anonymous Author(s)
Affiliation
Address
email

Abstract

Organizations allocating work among humans, AI systems, and human–AI teams must balance service, cost, limited review capacity, correlated failures, and hard operating requirements. Indiscriminate adoption of an uncertain computational technology creates a second trust problem: a promising candidate is not sufficient evidence that the technology should be invoked. Q-OPS treats quantum computation as an optional expert inside an application-oriented decision architecture. Strong classical methods establish a credible incumbent and characterize remaining headroom; a future pre-execution router may invoke an optional quantum pilot; the frozen post-simulation gate records evidence after that pilot; and quantum-independent checking by the shared classical evaluator preserves fallback. In the frozen study, the quantum branch audits a retained-state distribution and does not replace the incumbent. Among 32 synthetic held-out instances, 9/32 (28.13%) meet the recorded post-simulation evidence gate. Across all retained-core comparisons, the ratio of exact optimum masses under the $p = 2$ and uniform distributions is $6.831\times$; this is distinct from the matched 128-draw outcomes. The evaluator records zero disallowed-mode violations, while only 27/32 classical incumbents satisfy the stricter composite-feasibility predicate. Earlier C1.0–C1.1 development stages and the frozen depth comparison preserve negative evidence: strong classical controls can remove apparent headroom, and $p = 1$ exceeds $p = 2$ on 28/32 cases under unequal parameter-search budgets. The present result is therefore statevector-level conditional algorithmic evidence, not real-QPU, wall-clock, scaling, universal, security, QML, or application-level quantum advantage. Q-OPS illustrates a general rule for trustworthy emerging-computation adoption: compare against credible alternatives, invoke selectively, verify through a quantum-independent path, and retain a fallback.

1 Introduction

Future service organizations allocate work among humans, AI systems, and human–AI teams. Each choice changes expected service, cost, capacity, accountability, review burden, and exposure to correlated AI failures. Human–AI staffing is therefore an economic resource-allocation problem with hard operational and institutional constraints, not merely a scheduling toy. Research on human–AI interaction calls for explicit authority and oversight boundaries [1, 25]; service operations supplies capacity and reliability models [11]; and sociotechnical analysis cautions that efficiency or a compact fairness proxy cannot stand in for stakeholder welfare [26].

Quantum optimization introduces a second adoption decision. Even if a statevector experiment concentrates probability on good candidates, using the quantum component may be irrational once classical alternatives, validation, delay, review, and risk costs are counted. Small noiseless demonstrations also do not establish QPU performance, wall-clock advantage, or scaling [3, 5, 21]. Rigorous

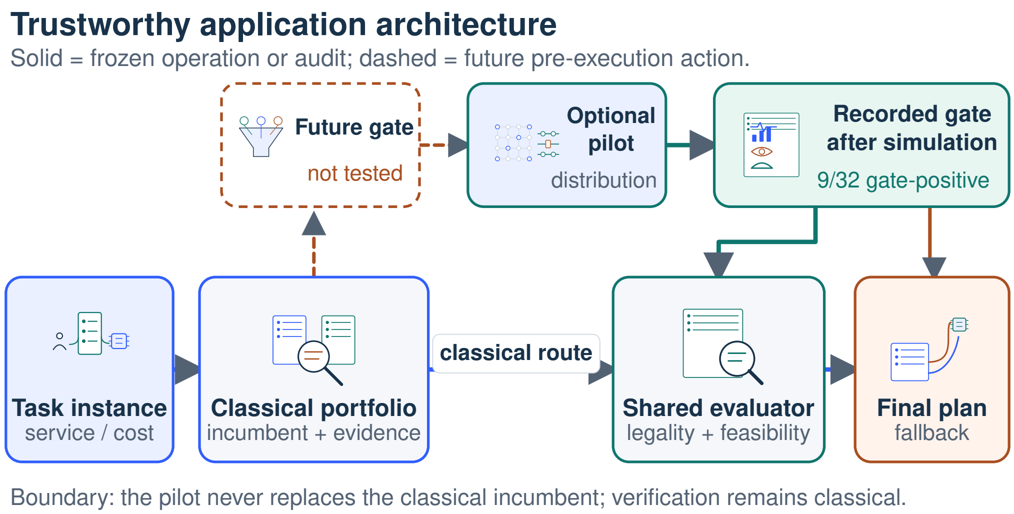

Figure 1: **Trustworthy application-level architecture.** The dominant solid lane produces and verifies a classical incumbent. A dashed future pre-execution gate may invoke an optional pilot; the separate frozen gate is computed only after simulation. Gate-positive candidates would return to the same classical evaluator, while unsupported use retains the incumbent. Clip art is semantic only; labels, arrows, and evidence states remain editable.

optimization benchmarks consequently emphasize standardized instances, strong counterfactuals, solution quality, resources, and reproducibility [16]; application-centric frameworks additionally ask whether the complete workflow improves a meaningful task [9].

Q-OPS moves the question one level upward: *when is there enough evidence to invoke quantum computation at all?* The architecture in Figure 1 treats quantum computation as an optional expert after a strong classical stage and before quantum-independent verification by the shared classical evaluator and fallback. It distinguishes system-level benchmarking (a fixed stack), algorithm-level benchmarking (controlled solver comparison), and application-level benchmarking (incremental decision value for the same organizational problem). The frozen study reaches the second level only on selectively retained cores inside an application-oriented architecture.

The paper is organized around exactly three research-question–contribution pairs:

RQ1. How can Human–AI staffing jointly represent service, cost, review scarcity, collaboration, correlated failures, and hard requirements? We contribute a synthetic, governance-constrained benchmark with human-only, AI-only, and collaborative modes, empirical chance constraints, disruption scenarios, and an explicit composite feasibility predicate.

RQ2. When should a trustworthy decision system remain classical or run a quantum pilot? We contribute an evidence-aware routing architecture driven by a multidimensional instance state, remaining classical headroom, ambiguity, feasibility, quantum-independent verification, and fallback. Abstention and human review are architectural actions, not evaluated policies in the frozen experiment.

RQ3. What conditional value survives strong counterfactuals and application-aware evidence boundaries? We contribute a frozen, null-inclusive evaluation with matched sampling budgets, negative development stages, exact shared-field reproduction, and an evidence ladder that separates sampling improvement from optimization, application, and welfare claims.

The headline result is deliberately two-sided: 9/32 cases satisfy a retrospective gate, while the all-instance retained-core comparison shows a 6.831× ratio of exact optimum masses under the frozen distributions. This is L2 statevector evidence, not a prospective router, matched-draw estimate, hardware/runtime advantage, or application benefit. The shared classical evaluator records zero disallowed-mode violations, but composite feasibility holds for only 27/32 classical incumbents;

unequal depth-search budgets prevent a depth claim. Trustworthy adoption therefore means justified, not maximal, quantum use.

2 Related Work and Human–AI Staffing Formulation

Intersecting literatures. Algorithm selection asks which solver fits an instance [17, 22]; selective prediction and learning to defer add abstention or handoff under uncertainty [12, 18]. QAOA and alternating operators provide parameterized distributions and constraint-aware mixers [8, 13], while warm starts inject classical information [7]. QOBLIB exemplifies model-independent instances, strong classical records, standardized metrics, and transparent resource reporting [16]. Q-OPS connects these components to a staffing decision in which verification and fallback remain classical. Its integration claim is qualified: it does not introduce a new circuit family, prove a welfare theorem, or establish field validity.

Selection, deferral, and evidence-aware invocation. Classical algorithm selection predicts which available solver will perform best on an instance [17, 22]. Selective prediction instead permits a model to withhold an answer, while learning to defer allocates a decision to another agent whose error profile may differ [12, 18]. Q-OPS borrows the option structure but not their empirical guarantees. Its recorded gate is computed after observing a quantum statistic; it is therefore a retrospective evidence partition, not a learned selector, calibrated uncertainty score, or prospective deferral policy. A gate evaluated only after paying the quantum cost cannot demonstrate cost-saving invocation.

Benchmark levels and credible counterfactuals. Quantum-optimization evidence depends on what is held fixed. System-level studies bind an algorithm to a particular software and hardware stack; algorithm-level studies compare methods under an explicit resource account; application-level studies ask whether inserting a quantum component improves the same decision problem relative to practically relevant alternatives. QOBLIB exemplifies strong classical records and standardized resource reporting [16], while application-benchmarking frameworks keep the surrounding workflow visible [9]. We synthesize these concerns into an evidence ladder rather than attributing an exact taxonomy to either source. The frozen comparison controls retained-state draws, but not parameter-search effort, total computation, hardware conditions, or organizational outcomes. Its contribution is consequently conditional algorithmic evidence inside an application-oriented architecture.

Human–AI allocation and institutional stakes. Human–AI work design is not exhausted by choosing the numerically best mode. Prior work distinguishes automation from augmentation and shows that their value depends on between-task and within-task complementarity [10]; interaction research emphasizes authority boundaries and institution-level consequences [1, 25]. Q-OPS encodes these concerns only as synthetic modes, capacities, disruption scenarios, and explicit diagnostics. It does not observe workers, customers, distributional effects, rights, or welfare. The contribution to trustworthy AI for social benefit is therefore a testable governance architecture and an explicit measurement agenda, not a claim of realized good.

Trustworthy quantum scope. The quantum trust question concerns a hybrid optimization pipeline: when a probabilistic quantum component should be invoked, how its candidate is checked without trusting the generation path, and how the system falls back when evidence is insufficient. The method is QAOA-based optimization, not quantum machine learning; it contains no learned quantum model, adversarial threat experiment, privacy mechanism, cryptographic claim, or real-QPU robustness result. Security and QML generalization are out of scope rather than silently implied by venue alignment.

Staffing benchmark. For tasks $i \in I$, choose a legal mode $y_i \in D_i \subseteq \{\mathrm{H}, \mathrm{A}, \mathrm{C}\}$ for human-only, AI-only, or collaboration. Tasks have duration d_i, priority p_i, value v_i, region $r(i)$, and mode-dependent cost c_{i,y_i}. Review-required AI work consumes one unit and collaboration one quarter,

$$C(y) = \sum_i d_i c_{i,y_i}, \qquad R(y) = \sum_i r_i \big(\mathbb{I}[y_i = \mathrm{A}] + .25\mathbb{I}[y_i = \mathrm{C}]\big), \tag{1}$$

with review overflow $R_+(y) = [R(y) - B_R]_+$. Workflow clusters receive a thresholded collaboration benefit, an AI-only concentration penalty, and a mode-friction penalty; the complete frozen formula appears in Appendix A.

Table 1: Benchmark and decision environment. All coefficients are frozen synthetic choices, not field-calibrated welfare weights.

Decision layer	Benchmark encoding	Trust question
Decision environment		
Decision	Human-only, AI-only, or human–AI collaboration	Accountability and automation choice
Reliability	24 correlated disruption scenarios; empirical chance constraints	Customer continuity under shocks
Oversight	Finite review capacity; collaboration uses partial review	Reviewer workload and escalation
Workflow	Cluster complementarity, concentration, and mode friction	Organizational coordination
Acceptance and evidence		
Objective	Expected service value minus cost, overflow, and penalties	Transparent trade-offs, not welfare
Acceptance	Composite feasibility, shared evaluator, classical fallback	Auditable operational boundary

For each correlated disruption scenario $\omega \in \Omega$, a shared evaluator computes weighted service and regional capacity overflow. Empirical service and capacity failure rates are

$$q_s(y) = |\Omega|^{-1}\sum_{\omega} \mathbb{I}[\text{service}_\omega(y) < .70], \quad q_c(y) = |\Omega|^{-1}\sum_{\omega} \mathbb{I}[O_\omega(y) > 0]. \tag{2}$$

Composite feasibility is $F(y) = \mathbb{I}[H(y) = 0, R_+(y) = 0, q_s(y) \leq .20, q_c(y) \leq .20]$, where H counts disallowed modes. The score combines expected value minus cost and overflow, workflow complementarity, a regional AI-exposure regularizer, review overflow, and chance-constraint penalties. All coefficients are frozen synthetic design choices, not learned welfare weights, workplace standards, or fairness guarantees.

The benchmark makes stakeholder consequences visible without claiming them as measured benefits. Customers depend on service continuity; workers face review load and inappropriate automation risk; organizations face cost, accountability, and resilience; society faces the broader question of disciplined technology adoption. Lower cost, more AI assignments, or more collaboration need not imply higher welfare.

3 Evidence-Aware Quantum–Classical Routing

Observable decision state. Let

$$z = [n, d, \rho, g, a, r, s, c] \tag{3}$$

summarize dimension, constraint density, interaction structure, remaining classical headroom, disagreement among strong classical methods, review scarcity, disruption severity, and computational cost. These components form an instance-difficulty vector rather than treating qubit count as difficulty. The frozen implementation observes only a subset directly; the complete vector is the intended application-level interface and its unevaluated components are future work.

Classical counterfactual and retained core. A legal coordinate-greedy solution is challenged by cluster destroy/repair with exact neighborhood enumeration. Let y^G, y^C be the incumbents, $\Delta = S(y^C) - S(y^G)$, d_G their normalized disagreement, and v_R restart-score variance. The frozen ambiguity score is

$$a = \min\left\{1,\ 2.8d_G + 5\left[\frac{\Delta}{\max(|S(y^C)|, 1)}\right]_+ + 4\frac{\sqrt{v_R}}{\max(|S(y^C)|, 1)}\right\}. \tag{4}$$

Four interacting tasks are retained and every legal assignment is classically evaluated with the incumbent fixed outside the core. Strictly feasible states form $\mathcal{Z}$; an empty set triggers an explicitly labeled least-violation fallback and never becomes "feasible." Exact enumeration therefore supplies a retained-core reference conditional on fixing the outside incumbent. Candidate legality and composite feasibility are checked separately by the shared scenario evaluator; neither operation certifies global optimality or field safety.

Optional quantum pilot. The warm state is the retained assignment nearest y^C. A feasible-subspace statevector QAOA distribution uses the retained-state adjacency as mixer and standardized score as cost. Uniform feasible sampling and QAOA receive the same 128-draw sampling budget. The empirical frozen gate requires a strict core, $a \geq .08$, classical headroom/disagreement, and $p_2^\star \geq \max(.18, 1.4p_U^\star)$. Since it consumes $p_2^\star$, it is a *post-simulation evidence gate*, not a deployable pre-execution policy.

The frozen quantum branch audits the distribution and does not replace the cluster-repair incumbent. The prospective policy is instead

$$\pi(z) = \begin{cases} \text{Quantum Pilot}, & \widehat{F}(z) = 1,\ H(z) > \tau_H,\ A(z) > \tau_A,\ \widehat{V}_Q(z) > \tau_Q, \\ \text{Classical}, & \text{strong classical evidence is sufficient}, \\ \text{Abstain / Human Review}, & \text{evidence is insufficient or risk is unacceptable}, \end{cases} \tag{5}$$

where $\widehat{F}(z)$ predicts pre-execution availability of an admissible retained state; it is not the realized assignment predicate $F(y)$. The frozen study neither estimates $\widehat{F}$ nor evaluates this prospective policy. Abstention/review are conceptual extensions. Economically, quantum invocation is justified only when

$$\mathbb{E}[\Delta W \mid z] > C_Q + C_{\text{validation}} + C_{\text{risk}} + C_{\text{review}}. \tag{6}$$

Q-OPS estimates none of these welfare terms; the inequality is a decision scaffold, not a theorem.

Failure model and containment. The architecture separates five failure modes that a single score would hide. A *false invocation* spends quantum, validation, and review resources when the classical incumbent was sufficient. A *false abstention* forgoes a genuinely useful candidate. A *candidate failure* returns a disallowed or composite-infeasible assignment. A *model failure* arises when the synthetic evaluator omits a stakeholder-relevant requirement or misprices a consequence. Finally, a *distribution failure* occurs when workload, capacity, or disruption behavior departs from the frozen generator. The current experiment addresses only a subset: the post-simulation gate exposes selective evidence; exact retained-state evaluation characterizes the local comparison; the shared evaluator rejects inadmissible candidates; and fallback preserves the cluster-repair incumbent. It does not estimate false-invocation cost, detect evaluator misspecification or distribution shift, or establish independence against common-mode software defects.

Verification is independent of the quantum proposal mechanism but uses the same classical evaluator as the benchmark, so it is not a third-party certificate or formal proof. A stronger implementation would separate proposal and verification code paths, version the decision model, log gate reasons and resource use, test verifier disagreement, and escalate unresolved cases to a named human authority. Those controls are prospective requirements, not properties measured by C1.2.

Verification and evidence ladder. Every candidate passes the same classical scenario evaluator before acceptance; otherwise the system returns the classical incumbent or escalates. We distinguish five evidence levels: L1 quantum feasibility; L2 conditional algorithmic value under controlled budgets; L3 hardware-level value; L4 end-to-end wall-clock value; and L5 application-level economic value. The present paper reaches L2 across frozen retained-core comparisons only; 9/32 cases satisfy the post-simulation evidence gate, but no prospective adoption policy was evaluated.

4 Experimental Design

The immutable C1.2 matrix crosses task counts $\{12, 18\}$, four synthetic stress levels, and held-out seeds $\{101, 211, 307, 401\}$ for 32 instances. Development seeds 11 and 23 are excluded. Each instance uses 24 correlated scenarios, three regions, three-task workflow clusters, service target .70, chance tolerance .20, review capacity, and a four-task retained core. QAOA depths $p = 1, 2$ are frozen; $p = 1$ receives a $13 \times 11 = 143$-point grid and $p = 2$ receives 30 seeded random trials, so depth comparisons are descriptive rather than an equal-optimizer-budget ablation.

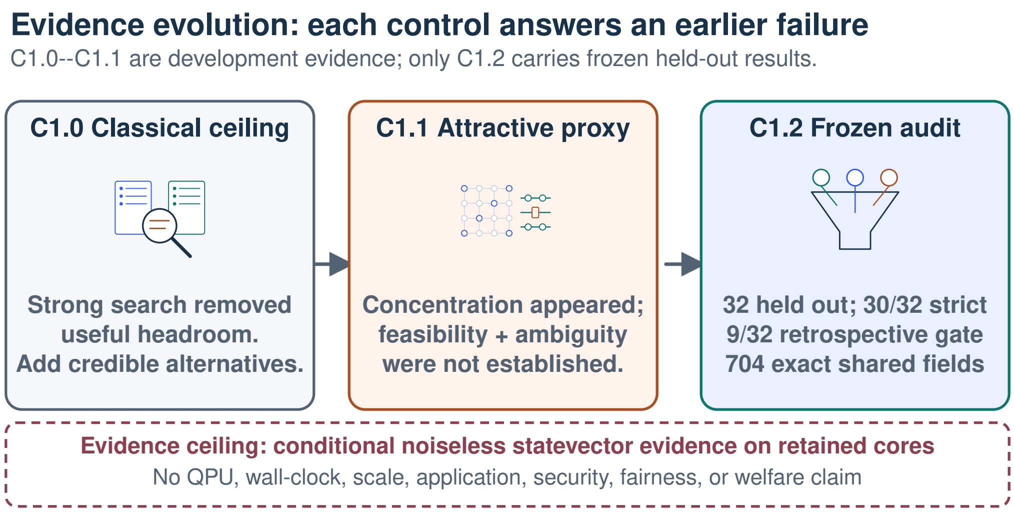

Figure 2: **Evidence evolution as a control argument.** C1.0 shows why credible classical alternatives are necessary; C1.1 shows why concentration alone is insufficient; C1.2 freezes strict feasibility, ambiguity/headroom, matched draws, a recorded gate, shared verification, and fallback. Only the immutable C1.2 held-out evaluation supplies headline numbers.

Uniform and $p = 2$ retained-core probabilities each receive 128 draws. Prespecified outcomes include strict-core availability, classical gain and ambiguity, optimum mass, first-hit draw (miss coded 129), near-optimal coverage, gate status, composite feasibility, and disallowed-mode violations. Percentile 95% bootstrap intervals resample paired instances 10,000 times. Sampling probability, best-of-budget assignment, practical runtime, application outcomes, and welfare remain distinct estimands.

Resource account and reproducibility boundary. Matched draws answer a narrow sampling question: given two retained-state distributions and 128 samples from each, how often is an optimum observed? They do not match the work required to construct the retained core, enumerate strict states, optimize QAOA parameters, simulate a statevector, communicate with hardware, verify candidates, or recover from failure. The $p = 1$ grid and $p = 2$ random search are also unequal optimization budgets. We therefore report sampling probability separately from best-of-budget assignment quality and exclude runtime, energy, and cost conclusions. The frozen configuration and null-inclusive development history implement a reproducibility discipline in which negative stages are evidence rather than discarded tuning attempts [20]. Reproduction here means executing the same code and data under documented conditions; it is not an independent replication on new instances or another implementation.

The development sequence is preserved rather than rewritten as a success-only narrative. C1.0 left too little useful headroom after strong classical search. C1.1 showed probability concentration without jointly establishing strict feasibility and classical ambiguity. C1.2 froze feasibility, headroom/ambiguity, matched retained-state comparison, selective gating, a quantum-independent classical verification boundary, and fallback before the held-out run. Figure 2 shows why each control is necessary; only C1.2 carries headline numbers.

A fresh runner produced 32 rows. Across the 22 shared fields and 32 records, all 704 comparisons match the archived table exactly (maximum absolute error 0). Eight archived diagnostics are no longer emitted, so the claim is exact *shared-field* reproduction, not byte-identical CSV reproduction. New router baselines, gate ablations, risk–coverage curves, QPU runs, and stronger full-problem solvers were not reproducible from the frozen protocol without changing the study; they are therefore preregistered evaluation priorities rather than fabricated results. The configuration digest and full environment record appear in Appendix C.

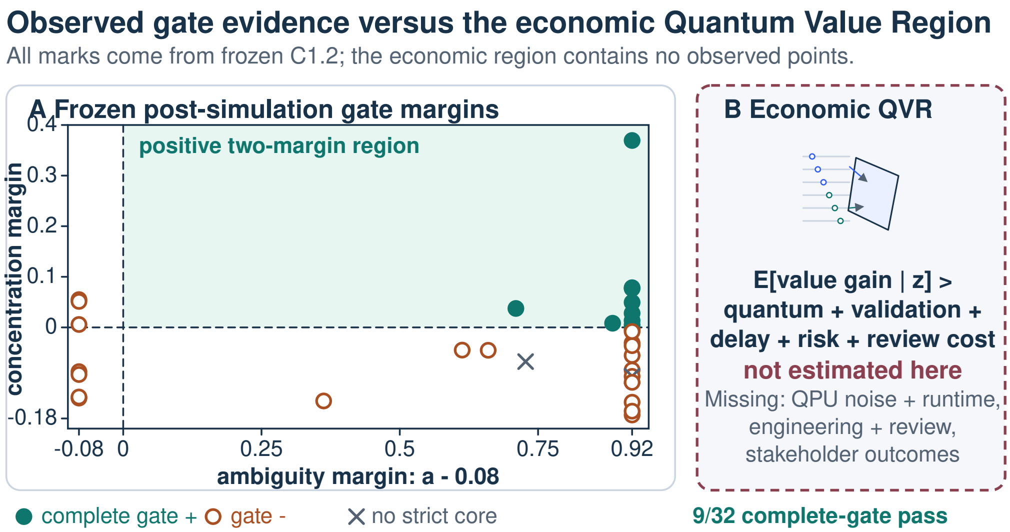

Figure 3: **Observed gate diagnostics and the unmeasured economic Quantum Value Region (QVR).** Panel A plots every frozen instance at its ambiguity margin $a - .08$ and concentration margin $p_2^\star - \max(.18, 1.4 p_U^\star)$. Filled points pass the complete post-simulation gate; open points fail another condition; crosses lack a strict core. Panel B separates the conceptual economic QVR, for which QPU, cost, runtime, risk, review, and stakeholder outcomes were not measured.

5 Results and Analysis

Classical structure and feasibility. Strict feasible retained states exist on 30/32 instances. Cluster repair improves over coordinate greedy on 25/32, with mean score gain 17.019 (paired bootstrap 95% CI [12.524,21.552]); collaboration rises descriptively from .475 to .664. These findings establish a meaningful classical counterfactual inside the frozen benchmark, but not superiority over mixed-integer programming, CP-SAT, exhaustive full-instance optimization, or a broad ALNS portfolio. Only 27/32 cluster incumbents satisfy composite feasibility even though all 32 record zero disallowed-mode violations, underscoring why verification and fallback cannot be collapsed into a single metric.

Selective conditional evidence. Nine of 32 cases (28.13%) pass the recorded gate. Across all 32 retained-core comparisons, mean optimum mass is .0223 for uniform and .1522 for $p = 2$, a ratio of 6.831× and paired difference .1299 (95% CI [.0991,.1640]); the first-hit proxy changes from 36.50 to 12.94 draws (paired difference -23.56 $[-39.28, -8.69]$). The near-optimal coverage difference is +.0443 $[-.0208, .1120]$, whose interval crosses zero. Figure 3 maps all observations: low two-solver ambiguity provides little recorded headroom signal, while positive ambiguity and concentration margins identify candidates subject to the remaining gates; neither region establishes economic value.

Negative and null evidence. The frozen ordered gate failures are two cases without a strict core, seven with low ambiguity, and fourteen with inadequate $p = 2$ concentration. The low gate-positive rate is a trust feature only in the limited sense that the recorded gate rejects universal use; because the gate is post-simulation, it does not reduce quantum calls. Mean $p = 1$ optimum mass is .2492 and $p = 1 > p = 2$ on 28/32 cases, but the parameter-search budgets differ, so neither deeper-is-better nor shallower-is-better is identified. Uniform feasible sampling is a controlled baseline, not a strong simulated-annealing, tabu, beam-search, or distributional classical portfolio.

What routing evaluation still requires. A leakage-free test must freeze a router using pre-execution features and compare Always Classical, Always Quantum, Random, Evidence-Aware, and a hindsight oracle on held-out instances. Coverage $N^{-1} \sum_i \mathbb{I}[\pi_\tau(z_i) = Q]$, conditional gain, compute/review cost, violations, verification, and fallback should be reported as thresholds vary. The current data cannot support that curve without retrofitting a prospective policy, so Q-OPS reports the test design and limitation rather than inventing points.

Table 2: Frozen C1.2 benchmark results. Brackets are 95% intervals only in rows that display them; the scope column distinguishes marginal and paired bootstrap summaries.

Quantity	Estimate [95% CI]	Evidence boundary
Held-out matrix	32	Frozen; all cases retained
Strict feasible core	30/32	2 least-violation fallbacks
Cluster-repair gain	17.019 [12.524, 21.552]	Paired instance bootstrap
Uniform optimum mass	0.0223 [0.0147, 0.0350]	Marginal instance bootstrap; exact distribution
$p = 2$ optimum mass	0.1522 [0.1161, 0.1940]	Marginal instance bootstrap; statevector
Ratio of exact means	6.831×	All 32 retained-core comparisons; no CI
$p = 2$ minus uniform	0.1299 [0.0991, 0.1640]	Paired instance bootstrap
Recorded gate positive	9/32 (28.13%) [12.50%, 43.75%]	Post-simulation; not prospective
$p = 1 > p = 2$	28/32	Unequal parameter-search budgets; descriptive
Composite-feasible incumbent	27/32	Stricter than mode legality
Disallowed-mode violations	0/32	Not total safety or feasibility

6 Trustworthiness, Limitations, and Conclusion

Trust mechanisms and human implications. The architecture exposes the classical counterfactual, records why a quantum branch is considered, checks every candidate with the shared classical evaluator independently of the quantum generation path, and preserves fallback. These mechanisms support auditability; they do not certify safe deployment. Customers need continuity under shocks, workers need limits on automation and review overload, and managers need accountable cost–service trade-offs. Society benefits only if adoption standards prevent performance proxies from displacing rights, welfare, and institutional legitimacy. Lower staffing cost is not automatically greater welfare; more AI or collaboration is not automatically desirable.

The stakeholder pathway is deliberately conditional. If a prospective router reduced unnecessary computation while preserving service and feasibility, organizations could redirect scarce validation and review capacity; if it failed, it could instead increase delay, opacity, and technology dependence. Workers could benefit from protected human-mandatory modes and explicit review limits, yet the same abstraction could conceal task quality, autonomy, surveillance, or unequal burden. Customers could benefit from continuity under correlated failures, but the synthetic service score is not customer welfare. These opposing pathways define what a trustworthy-AI-for-good evaluation must measure with affected stakeholders before deployment.

The same boundary sharpens the trustworthy-quantum claim. The present controls address evidence logging, candidate checking, and fallback in an idealized optimizer. They do not address malicious inputs, privacy leakage, supply-chain compromise, verifier attacks, or noise-induced QPU failures. Those are distinct security and robustness questions. Treating them as already solved would turn architectural intent into an unsupported assurance claim.

Scientific limits. The benchmark is synthetic: costs, success rates, disruptions, and penalties are not estimated from organizations or affected workers. There are no human subjects or personal data. Regional AI exposure is not fairness, collaboration is not worker benefit, and zero disallowed-mode violations is not workplace safety. Quantum evidence is limited to four retained ternary decisions, exact enumeration, noiseless statevectors, unequal depth-search budgets, no circuit decomposition, no QPU, no noise, no wall-clock or energy accounting, and no scaling test. Stronger classical full-problem and retained-core samplers could reduce or eliminate the observed signal. Bootstrap intervals describe these 32 frozen instances, not a population of workplaces.

Application-level path. Progress from L2 to L5 requires a prospective router, equal and widened classical budgets, hardware/noise studies, end-to-end time/cost accounting, operational outcomes, stakeholder-defined measures, and field-valid data with governance. Any future deployment needs human override, privacy controls, distribution-shift monitoring, and a documented accountability owner. NIST and ISO frameworks are governance references, not certifications [14, 15, 27]. The supported conclusion is justified, not maximal, quantum use.

References

[1] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. Guidelines for human–AI interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, pages 1–13, 2019. doi: 10.1145/3290605.3300233.

[2] Bin Bai, Hengchen Dai, Dennis J. Zhang, Fanyin Zhang, and Hao Hu. The impacts of algorithmic work assignment on fairness perceptions and productivity: Evidence from field experiments. *Manufacturing & Service Operations Management*, 24(6):3060–3078, 2022. doi: 10.1287/msom.2022.1120.

[3] Kishor Bharti, Alba Cervera-Lierta, Thi Ha Kyaw, Tobias Haug, Sumner Alperin-Lea, Abhinav Anand, Matthias Degroote, Hermanni Heimonen, Jakob S. Kottmann, Tim Menke, Wai-Keong Mok, Sukin Sim, Leong-Chuan Kwek, and Alán Aspuru-Guzik. Noisy intermediate-scale quantum algorithms. *Reviews of Modern Physics*, 94:015004, 2022. doi: 10.1103/RevModPhys.94.015004.

[4] Kostas Blekos, Dean Brand, Andrea Ceschini, Chiao-Hui Chou, Rui-Hao Li, Komal Pandya, and Alessandro Summer. A review on quantum approximate optimization algorithm and its variants. *Physics Reports*, 1068:1–66, 2024. doi: 10.1016/j.physrep.2024.03.002.

[5] Marco Cerezo, Andrew Arrasmith, Ryan Babbush, Simon C. Benjamin, Suguru Endo, Keisuke Fujii, Jarrod R. McClean, Kosuke Mitarai, Xiao Yuan, Lukasz Cincio, and Patrick J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3:625–644, 2021. doi: 10.1038/s42254-021-00348-9.

[6] Abraham Charnes and William W. Cooper. Chance-constrained programming. *Management Science*, 6(1):73–79, 1959. doi: 10.1287/mnsc.6.1.73.

[7] Daniel J. Egger, Jakub Mareček, and Stefan Woerner. Warm-starting quantum optimization. *Quantum*, 5:479, 2021. doi: 10.22331/q-2021-06-17-479.

[8] Edward Farhi, Jeffrey Goldstone, and Sam Gutmann. A quantum approximate optimization algorithm. *arXiv preprint arXiv:1411.4028*, 2014. doi: 10.48550/arXiv.1411.4028.

[9] Jernej Rudi Finžgar, Philipp Ross, Leonhard Hölscher, Johannes Klepsch, and Andre Luckow. QUARK: A framework for quantum computing application benchmarking. In *2022 IEEE International Conference on Quantum Computing and Engineering*, pages 226–237, 2022. doi: 10.1109/QCE53715.2022.00042.

[10] Andreas Fügener, Dominik D. Walzner, and Alok Gupta. Roles of artificial intelligence in collaboration with humans: Automation, augmentation, and the future of work. *Management Science*, 72(1):538–557, 2026. doi: 10.1287/mnsc.2024.05684.

[11] Noah Gans, Ger Koole, and Avishai Mandelbaum. Telephone call centers: Tutorial, review, and research prospects. *Manufacturing & Service Operations Management*, 5(2):79–141, 2003. doi: 10.1287/msom.5.2.79.16071.

[12] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[13] Stuart Hadfield, Zhihui Wang, Bryan O'Gorman, Eleanor G. Rieffel, Davide Venturelli, and Rupak Biswas. From the quantum approximate optimization algorithm to a quantum alternating operator ansatz. *Algorithms*, 12(2):34, 2019. doi: 10.3390/a12020034.

[14] International Organization for Standardization. ISO/IEC 23894:2023: Information technology—artificial intelligence—guidance on risk management. Technical report, ISO, 2023. URL `https://www.iso.org/standard/77304.html`.

[15] International Organization for Standardization. ISO/IEC 42001:2023: Information technology—artificial intelligence—management system. Technical report, ISO, 2023. URL `https://www.iso.org/standard/42001.html`.

[16] Thorsten Koch, David E. Bernal Neira, Ying Chen, Giorgio Cortiana, Daniel J. Egger, Raoul Heese, Narendra N. Hegade, Alejandro Gomez Cadavid, Rhea Huang, Toshinari Itoko, Thomas Kleinert, et al. The quantum optimization benchmarking library. *Nature Computational Science*, 6:653–671, 2026. doi: 10.1038/s43588-026-00991-1.

[17] Lars Kotthoff. Algorithm selection for combinatorial search problems: A survey. *AI Magazine*, 35(3):48–60, 2014. doi: 10.1609/aimag.v35i3.2460.

[18] David Madras, Toni Pitassi, and Richard Zemel. Predict responsibly: Improving fairness and accuracy by learning to defer. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[19] Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6):1–35, 2021. doi: 10.1145/3457607.

[20] Joelle Pineau, Philippe Vincent-Lamarre, Koustuv Sinha, Vincent Larivière, Alina Beygelzimer, Florence d'Alché Buc, Emily Fox, and Hugo Larochelle. Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program. *Journal of Machine Learning Research*, 22(164):1–20, 2021.

[21] John Preskill. Quantum computing in the NISQ era and beyond. *Quantum*, 2:79, 2018. doi: 10.22331/q-2018-08-06-79.

[22] John R. Rice. The algorithm selection problem. *Advances in Computers*, 15:65–118, 1976. doi: 10.1016/S0065-2458(08)60520-3.

[23] Stefan Ropke and David Pisinger. An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows. *Transportation Science*, 40(4):455–472, 2006. doi: 10.1287/trsc.1050.0135.

[24] Nicolas P. D. Sawaya, Alexander T. Schmitz, and Stuart Hadfield. Encoding trade-offs and design toolkits in quantum algorithms for discrete optimization: Coloring, routing, scheduling, and other problems. *Quantum*, 7:1111, 2023. doi: 10.22331/q-2023-09-14-1111.

[25] Isabella Seeber, Eva Bittner, Robert O. Briggs, Triparna de Vreede, Gert-Jan de Vreede, Aaron Elkins, Ronald Maier, Alexander B. Merz, Sarah Oeste-Reiß, Nils Randrup, Gerhard Schwabe, and Matthias Söllner. Machines as teammates: A research agenda on AI in team collaboration. *Information & Management*, 57(2):103174, 2020. doi: 10.1016/j.im.2019.103174.

[26] Andrew D. Selbst, danah boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. Fairness and abstraction in sociotechnical systems. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, pages 59–68, 2019. doi: 10.1145/3287560.3287598.

[27] Elham Tabassi. Artificial intelligence risk management framework (AI RMF 1.0). Technical Report NIST AI 100-1, National Institute of Standards and Technology, U.S. Department of Commerce, 2023.

A Benchmark formulas, meanings, and precedents

This appendix distinguishes four evidence statuses: *implemented* means the public evaluator computes the quantity; *synthetic* means its coefficient is generated or frozen rather than estimated from field data; *interpretive* means we supply a real-world analogy; and *validated* is deliberately not asserted for organizational welfare, safety, or fairness.

For scenario ω, let D_ω be demand, $h_{r\omega}$ and $a_{r\omega}$ regional human/AI factors, and u_ω the common AI shock. With clipped success to $[0, 1]$,

$$s_{i\omega}(\mathrm{H}) = \bar{s}_i^H h_{r(i)\omega}, \qquad (\ell^H, \ell^A)(\mathrm{H}) = (1, 0), \tag{7}$$

$$s_{i\omega}(\mathrm{A}) = \bar{s}_i^A a_{r(i)\omega} u_\omega, \qquad (\ell^H, \ell^A)(\mathrm{A}) = (0, 1), \tag{8}$$

$$s_{i\omega}(\mathrm{C}) = \bar{s}_i^C (.55 h_{r(i)\omega} + .45 a_{r(i)\omega})(.985 + .015 u_\omega), \qquad (\ell^H, \ell^A)(\mathrm{C}) = (.45, .65). \tag{9}$$

Regional overflow is the sum of positive excesses over scenario-scaled human and AI capacities:

$$O_\omega(y) = \sum_r \left[L^H_{r\omega}(y) - B^H_r h_{r\omega}\right]_+ + \left[L^A_{r\omega}(y) - B^A_r a_{r\omega}\right]_+, \tag{10}$$

where $L^x_{r\omega} = \sum_{i:r(i)=r} d_i D_\omega \ell^x(y_i)$. These equations correct a tempting but inaccurate simplification: the service constraint is aggregate and scenario-based, not a separate chance constraint for every task.

Table 3: Complete benchmark formula glossary. Every row is implemented in the frozen evaluator.

Factor	Implemented definition	Plain-language / real-world role	Evidence and literature
Mode	$y_i \in \{H, A, C\}$; $A \notin D_i$ for human-mandatory i	Each task is human-only, AI-only, or collaborative; some retain human responsibility.	Directly enumerated domain; human–AI interaction design motivates explicit authority boundaries [1, 25].
Hard violations	$H(y) = \sum_i \mathbf{1}[y_i \notin D_i]$	Counts disallowed assignments; this is narrower than composite feasibility.	Shared final check; governance context [15, 27].
Base cost	$C(y) = \sum_i d_i c_{i,y_i}$	Duration-weighted cost of the chosen operating mode.	Scenario-independent term in the evaluator; resource allocation tradition [11].
Review load	$R(y) = \sum_i r_i(\mathbf{1}[y_i = A] + .25\mathbf{1}[y_i = C])$	AI-only work consumes a full review unit; collaboration consumes one quarter.	Synthetic coefficient, not empirically calibrated; makes oversight capacity explicit [27].
Review overflow	$R_+(y) = [R(y) - B_R]_+$	Review demand above the available audit budget.	Penalized by 9 in the score; operational bottleneck construct.
Cluster reward	$\sum_k B_k \mathbf{1}[n^C_k \geq h_k]$	Collaboration creates value only after a workflow-level threshold.	Implemented clusters of at most three tasks; complementarity motivation [10, 25].
All-AI penalty	$\sum_k A_k \mathbf{1}[\forall i \in k : y_i = A]$	A fully automated workflow incurs a governance/robustness cost.	Synthetic penalty, increasing with stress level; not a causal harm estimate.
Handoff penalty	$\sum_k J_k(\lvert\{y_i : i \in k\}\rvert - 1)_+$	More modes inside one cluster create coordination overhead.	Implemented synthetic workflow-mixing cost; the coefficient is not empirically calibrated.
AI exposure dispersion	$P_{\exp}(y) = 2\,\mathrm{sd}_r(a_r)$, $a_r = \lvert I_r\rvert^{-1} \sum_{i\in I_r} \mathbf{1}[y_i = A]$	Penalizes uneven AI-only rates across modeled regions.	Coded diagnostic, *not* validated fairness; sociotechnical cautions [2, 19, 26].
Scenario loads	$L^H_{r\omega} = \sum_{i\in r} d_i D_\omega \ell^H(y_i)$, $L^A_{r\omega} = \sum_{i\in r} d_i D_\omega \ell^A(y_i)$, with $(\ell^H, \ell^A) = (1, 0), (0, 1), (.45, .65)$	Human and AI capacity draw changes by mode and correlated demand scenario.	Exact frozen coefficients; chance-constrained operations motivation [6, 11].
Scenario success	$s_{i\omega} = \bar{s}_{i,y_i}$ times regional human/AI factors and AI shock; collaboration blends factors $.55/.45$	Mode success changes with regional and common disruptions.	Exact generator/evaluator equations in Appendix A; synthetic, not field-estimated.
Service risk	$q_s(y) = \lvert\Omega\rvert^{-1} \sum_\omega \mathbf{1}[\rho_\omega(y) < .70]$, with $\rho_\omega = \sum_i p_i s_{i\omega} / \sum_i p_i$	Fraction of scenarios whose priority-weighted service falls below target.	Target .70, tolerance $\epsilon = .20$; empirical chance constraint [6].
Capacity risk	$q_c(y) = \lvert\Omega\rvert^{-1} \sum_\omega \mathbf{1}[O_\omega(y) > 0]$	Fraction of scenarios with any regional human/AI capacity overflow.	Tolerance $\epsilon = .20$; overflow also priced inside scenario value.
Scenario value	$V_\omega(y) = \sum_i v_i s_{i\omega} - C(y) - 4O_\omega(y)$	Expected delivered value net of cost and scenario overflow.	Averaged over 24 scenarios.

Factor	Implemented definition	Plain-language / real-world role	Evidence and literature
Composite score	$S(y) = \bar{V} +$ cluster reward $-$ cluster penalties $- P_{\text{exp}} - 9R_+ - 600H - 150[q_s - .2]_+ - 45[q_c - .2]_+$	A single score ranks assignments while retaining explicit diagnostics.	Exact frozen implementation; coefficients are benchmark parameters, not learned welfare weights.
Composite feasible	$F(y) = \mathbf{1}[H = 0, R_+ = 0, q_s \leq .2, q_c \leq .2]$	Requires mode legality, review capacity, service reliability and capacity reliability.	Used for strict retained states and reported separately from hard-mode legality.
Collaboration rate	$n^{-1} \sum_i \mathbf{1}[y_i = C]$	Describes how often the collaborative mode is selected.	Descriptive outcome only; it is not itself a success metric.

Generator distributions. The generator assigns three-task clusters, up to three modeled regions, task values in $[9, 18]$, human success in $[.94, .99]$, AI success based on $[.84, .955]$ with a sensitivity decrement, and collaborative success above the better base mode by a sampled margin, capped at .997. Costs, capacities, review requirements, cluster coefficients and disruption strength are seeded. The four stress levels increase sensitive-task prevalence, outage probability and selected penalties. These ranges define a controlled stress test, not an empirical model of any named industry.

B Complete U–W–C–Q formula glossary

The mnemonic is not execution order. The classical layer encloses the experiment: it produces the incumbent and retained set, while the same evaluator verifies the final assignment independently of the quantum generation path. The warm layer feeds Q, and U/Q form a paired distributional comparison. Figure 1 visualizes the enclosing application architecture; Table 4 records every implemented strategy and metric.

Table 4: Complete U–W–C–Q strategy and metric glossary.

Object	Formula / implementation	Intuitive definition	Use and precedent
Individual seed	$\arg\max_{m \in D_i}(v_i \bar{s}_{im} - d_i c_{im})$ per task	Myopic value-minus-cost initialization.	Starting point for coordinate descent.
Greedy control	Repeated best legal one-task move until no S improvement	Stronger than an all-human or random straw baseline.	Classical control; local-search precedent.
Cluster repair	Exact enumeration of one three-task cluster, sometimes plus one external task, for 18 iterations and 2 restarts	Destroy/repair coordinated assignments inside a small neighborhood.	Inspired by ALNS [23]; the implementation is intentionally reported as *cluster repair*, not a comprehensive ALNS suite.
Classical gain	$\Delta = S(y^C) - S(y^G)$	Improvement of cluster repair over greedy.	One ambiguity signal; mean and win rate are frozen outcomes.
Disagreement	$d = n^{-1} \sum_i \mathbf{1}[y_i^G \neq y_i^C]$	How differently the two classical procedures allocate tasks.	A second ambiguity signal.
Ambiguity	$a = \min\{1, 2.8d + 5[\Delta / \max(\lvert S_C \rvert, 1)]_+ + 4\sqrt{v_R} / \max(\lvert S_C \rvert, 1)\}$	Normalized evidence that the classical landscape is contested.	Threshold .08; diagnostic, not calibrated probability. Algorithm selection context [17, 22].
Core score	$B_k + A_k + J_k \max(1, \lvert k \rvert - 1) + 5\bar{d}_k$	Ranks clusters by complementarity, penalties and disagreement.	Selects four task indices; not a learned selector.
Retained set	Enumerate $\prod_{i \in I} D_i$; keep $F(y) = 1$	Exact small-core assignments that remain feasible with the incumbent outside the core.	At most $3^4 = 81$ states.

Object	Formula / implementation	Intuitive definition	Use and precedent
Fallback set	If no strict state, retain lowest 20% of $20H + 2R_+ + q_s + q_c$	Keeps a least-violating diagnostic set when strict feasibility fails.	Used on 2/32 instances; never relabeled feasible.
Retained graph	$A_{zz'} = 1$ for one-mode Hamming neighbors; connect isolated nodes to nearest retained state	Mixer moves only among retained states.	Feasible alternating-operator motivation [13, 24].
Warm state	$z_W = \arg\min_{z \in \mathcal{Z}} d_H(z, y_I^C)$; $\psi_0(z) \propto e^{-.35 d_H(z, z_W)}$	Starts near the cluster-repair incumbent while retaining support elsewhere.	Warm-start motivation [7].
Hamiltonians	$H_C = \mathrm{diag}((E - \bar{E})/\sigma_E)$, $E = -S$; $H_M = -A$	Phase by standardized cost; mix along retained edges.	Statevector implementation; no gate compilation or noise model.
QAOA state	$\lvert\psi_p\rangle = \prod_{\ell=1}^{p} U_M(\beta_\ell) U_C(\gamma_\ell) \lvert\psi_0\rangle$	Alternates mixer and cost evolution, with $U_X(t) = e^{-itH_X}$.	QAOA/alternating-operator line [4, 8, 13].
Parameter search	$p = 1$: 13×11 grid; $p = 2$: 30 seeded random trials; lexicographic objective $(\mathbb{E}[E], -M_{.05}, -p^\star)$	Frozen but unequal searches at the two depths.	Supports reporting each depth, not a causal depth comparison.
Uniform baseline	$P_U(z) = 1/\lvert\mathcal{Z}\rvert$	Uninformed sampling on the same retained set.	Isolates concentration from feasibility filtering.
Optimum mass	$p^\star = \sum_{z: E(z) = E_{\min}} P(z)$	Probability of drawing any exactly optimal retained state.	Primary statevector concentration metric.
Near-optimal mass	$M_{.05} = \sum_{z \in \mathcal{N}_{.05}} P(z)$, $\mathcal{N}_{.05} = \{E \leq E_{\min} + .05\,\mathrm{range}(E)\}$	Probability inside a 5%-of-range energy band.	Secondary optimizer tie-break and descriptive metric.
First hit	$T = \min\{b : z_b \text{ optimal}\}$; $T = 129$ on a miss, budget 128	Draw index of the first optimum under one seeded matched sample.	Sample-efficiency proxy, not wall-clock runtime.
Coverage	distinct sampled near-optimal states / all near-optimal retained states	Diversity inside the near-optimal band.	Paired difference interval crosses zero.
Evidence gate	$F_I \wedge a \geq .08 \wedge p_2^\star \geq \max(.18, 1.4 p_U^\star) \wedge (\Delta > 0 \vee d \geq .05)$	Flags frozen instances with strict feasibility, ambiguity and sufficient simulated concentration.	Computed *after* QAOA; not an invocation policy.
Verification	Re-evaluate the accepted full assignment with the shared scenario evaluator	Separates quantum evidence from operational acceptance without implying a third-party certificate.	In the current experiment the cluster-repair incumbent remains the final/fallback assignment.

Parameter-selection objective. For every frozen parameter candidate, statevector probabilities are ranked lexicographically by expected energy, then negative near-optimal mass, then negative exact-optimum mass. Thus $p^\star$ is not the sole optimization objective. The $p = 1$ grid and $p = 2$ random search also have unequal cardinality. Reporting the strong $p = 1$ result is therefore part of the audit rather than an ablation.

No production quantum assignment. The implementation computes an exact best retained-core state for reference and distributional sampling metrics, but it does not replace the cluster-repair incumbent with a sampled quantum candidate. "Quantum-independent verification" means that any accepted full assignment is evaluated by the shared classical scenario evaluator; it is not a third-party certificate or a separate implementation. In the frozen run, the operational incumbent remains classical. This distinction prevents probability concentration from being misreported as an implemented end-to-end policy improvement.

Table 5: Frozen experimental protocol.

Component	Frozen value
Factorial matrix	$n \in \{12, 18\}$; stress/doping $\in \{0, 1, 2, 3\}$; seeds $\{101, 211, 307, 401\}$; 32 instances total
Scenarios	24 correlated scenarios per instance; service target .70; tolerance .20; capacity scale .92
Classical	Coordinate greedy; cluster destroy/repair, 18 iterations, 2 restarts
Retained core	4 variables; ternary legal domains; exact retained-core enumeration; at most 81 states
Quantum	Noiseless feasible-subspace statevector; $p = 1, 2$; $p = 1$ grid 143 points; $p = 2$ seeded random search 30 points
Sampling	128 replacement draws per U/Q distribution; different frozen seeds; miss coded 129
Inference	Instance is the paired unit; percentile paired bootstrap; 10,000 resamples; all prespecified nulls retained
Config digest	`7f07459abc9e686934be1ee022768eb82` `a9687137cc86d4d7ea0b7727715e53c`
Source snapshot	Anonymous frozen implementation artifact; identifying commit retained in the external audit only

C Frozen protocol and pseudocode

Locked runner. For each tuple $(n, \text{stress}, \text{seed})$ in the frozen matrix:

1. generate the synthetic tasks, clusters, capacities and 24 correlated scenarios;
2. run coordinate greedy and cluster repair, preserving scores, assignments and restart variance;
3. rank clusters, select four task indices, exactly enumerate their legal assignments, and retain composite-feasible states (or explicitly marked least-violating fallback states);
4. compute the uniform distribution, the warm state, and optimized noiseless QAOA distributions at $p = 1, 2$;
5. draw 128 samples from uniform and $p = 2$ with frozen method-specific seeds;
6. calculate ambiguity and the post-simulation evidence gate; and
7. serialize every instance, including null and gate-negative cases, before aggregate bootstrap inference.

Inference. The instance is the paired unit. For a contrast d_j over $j = 1, \ldots, 32$, the reported percentile interval is the 2.5th and 97.5th percentiles of 10,000 means formed by resampling instance indices with replacement. Seeds 7001–7004 are fixed for gain, optimum-mass difference, first-hit difference and coverage difference. These intervals quantify the frozen matrix, not a random sample of real organizations.

Compute. The independent verification used Python 3.12.13 and NumPy 2.3.5 on Linux 6.18.35, an AMD EPYC 9V74 allocation with 9 logical CPUs and 21 GiB RAM. The complete 32-instance locked rerun took 31.980 seconds wall-clock and no GPU. Runtime is an environment/load observation, not an advantage metric. The archived manifest records Python 3.13.5 and NumPy 2.3.5 but not original CPU or wall time; this missing historical metadata is disclosed rather than reconstructed.

D Additional results and diagnostic partitions

Depth and matched draws. Across instances, mean optimum masses are .022278 (uniform), .249182 ($p = 1$), and .152190 ($p = 2$). The $p = 2$ value exceeds uniform on 30/32; $p = 1$ exceeds $p = 2$ on 28/32. Since the parameter searches are unequal and selected by expected energy before optimum mass, this is a warning against interpreting depth as the sole causal factor.

Uniform and $p = 2$ near-optimal coverage means are .919271 and .963542. Their paired mean difference .044271 has 95% interval $[-.020833, .111979]$. We therefore report the direction but

Table 6: Ordered post-simulation evidence-gate partition.

First failing/succeeding condition	Count	Interpretation
No strict feasible retained state	2	Violation-ranked fallback; cannot support strict-core evidence.
Classical ambiguity below .08	7	Strong classical methods give little reason to focus on disagreement.
Concentration below frozen threshold	14	QAOA $p = 2$ does not meet both absolute and relative mass thresholds.
All evidence conditions pass	9	Post-simulation evidence-positive subset; still not a deployment decision.

make no positive coverage claim. The first-hit means of 36.50 and 12.9375 are sensitive to the frozen draw stream and miss code; they complement, rather than replace, the exact probability-mass result.

Feasibility decomposition. Thirty instances have at least one strict feasible retained state; two use fallback states. All 32 cluster-repair incumbents have zero explicit disallowed-mode violations, while 27 satisfy the stricter conjunction of review capacity and empirical service/capacity tolerances. Any abstract phrase "zero final hard violations" should be read as zero mode-domain violations only.

Gate interpretation. The ordered partition in Table 6 prevents double counting. Its 9 positive instances meet all frozen predicates after QAOA. A real invocation router would have to replace $p_2^\star$ with pre-execution features and be evaluated on new held-out instances. Until then, "gate rate" is an evidence summary, not a rate of saved QPU calls.

E Reproducibility and claim traceability

The supplementary source contains the frozen configuration, archived 32-row evidence table, aggregate summary, rerun summary, configuration digest, figure generator, and a shared-field comparison script. For anonymous review, the public location is represented as the anonymous supplementary artifact; the camera-ready source should replace this placeholder with the permanent repository URL and commit.

Minimal reproduction. From the public source snapshot, install NumPy and pytest, set the repository root on `PYTHONPATH`, and run the three targeted test modules for the C1.1/C1.2 generator, evaluator and locked protocol. Then execute the locked runner with its explicit confirmation flag and a new empty output directory. Finally run `scripts/audit_reproduction.py` from this package against the archived and fresh CSVs. The independent audit obtained 32/32 rows, 704/704 exact shared-field matches and maximum absolute error zero.

The archived CSV has 30 columns; the current public runner emits 22. The eight archive-only diagnostics are restart variance, composite incumbent feasibility, retained-state count, fallback use, $p = 2$ near-optimal mass, effective support, and uniform/QAOA sample diversity. Because the schemas differ, the package does not claim byte identity. The richer archive remains the source for composite-feasibility and depth diagnostics; every field currently emitted by the runner is exactly reproduced.

Table 7: Claim-to-artifact traceability.

Claim / object	Public artifact or function	Verification route
Generator and evaluator	`qops/c12_generator.py`; `generate_instance`; `evaluate`	Exact formulas summarized in Appendix A; targeted tests cover determinism and legality.
Classical controls	`qops/c12_classical.py`; coordinate greedy; cluster repair	Rerun all 32 frozen instances; compare gain, disagreement and feasibility fields.

Claim / object	Public artifact or function	Verification route
Core and statevector	`qops/c12_quantum.py`; core extraction; retained-state QAOA	Recompute retained states, $p = 1/p = 2$ masses and matched draws.
Frozen matrix and inference	`qops/c12_locked.py`; locked config JSON	Explicit confirmation flag; configuration SHA-256; deterministic seeds.
Headline aggregates	Archived records CSV and summary JSON	32 archived rows; fresh run matches 704/704 values shared with the current runner exactly.
Figures and tables	`scripts/generate_release_assets.py` and copied frozen CSV	Regenerate live-text SVG masters and table inputs; export PDF; run structural and numerical audits.

Repository-wide health caveat. The targeted frozen experiment passes. A broad legacy test discovery also surfaced an unrelated sparse-tracking certificate failure, and a clean environment initially lacked pytest for a protocol module. These do not alter the C1.2 values, but they mean the snapshot should not be described as having an entirely green repository-wide test suite. The independent report records this distinction.

F Expanded limitations, ethics, and release guidance

Construct validity. The benchmark combines factors that are individually plausible but jointly synthetic. Penalty weights trade unlike quantities inside one score; they are not moral exchange rates. Regional AI exposure is observed at the region level and omits protected attributes, within-region heterogeneity, job quality, worker consent and downstream customer outcomes. The framework should therefore be used to test algorithms, not to rank actual employees or justify automation.

Quantum validity. Exact enumeration of at most 81 retained states makes ground truth available and is a virtue for auditing, but also removes the scale at which computational advantage could arise. Statevector probability ignores hardware sampling noise, connectivity, compilation, calibration drift, queue time and energy. Warm initialization and a retained graph are simulated as dense linear algebra; their circuit cost is not estimated. A hardware study must report qubits, gates, depth, shots, error mitigation, compilation, wall-clock and comparable classical compute.

Classical validity. Coordinate greedy and cluster repair are meaningful controls but not optimality certificates for the full 12/18-task problem. Exact enumeration supplies only a retained-core reference conditional on the outside incumbent. Future work should add mixed-integer, constraint-programming and stronger heuristic portfolios with equal tuning budgets.

Potential benefits and harms. Explicit abstention, audit burden and human-mandatory modes could support accountable system design. The same optimizer could also rationalize labor substitution, centralize authority, or disguise a contested value judgment as a technical coefficient. Mitigations include participatory specification, disaggregated outcome reporting, review-capacity stress tests, appeal/override channels, privacy protection and independent impact assessment. Standards such as NIST AI RMF and ISO/IEC 42001/23894 provide governance vocabulary, but compliance and conformity assessment are outside this study.

Anonymous and camera-ready artifacts. The compiled submission omits an identifying URL to preserve double blindness. Before camera-ready release, authors should insert the permanent public URL and commit, verify its license, archive the exact artifact, and ensure the paper does not imply that governance references endorse the benchmark.

NeurIPS Paper Checklist

1. **Claims**

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract, Introduction, Results, and Table 2 state the conditional statevector claim and explicitly exclude QPU, runtime, scaling, safety, and universal-advantage claims.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 6 and Appendix F discuss synthetic construct validity, classical controls, small retained cores, unequal searches, noiseless simulation, and societal risks.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper introduces an executable benchmark and empirical audit, not new theorems or formal proofs; all implemented formulas and assumptions are nevertheless disclosed.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.

- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Sections 4 and Appendices C–E specify the frozen matrix, algorithms, seeds, metrics, inference, environment, digest, commands, and exact shared-field audit.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
     (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The public MIT-licensed implementation and frozen outputs are available through the anonymous supplementary artifact during review; Appendix E describes access and reproduction.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (https://neurips.cc/public/guides/CodeSubmissionPolicy) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (https://neurips.cc/public/guides/CodeSubmissionPolicy) for more details.

- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 4, Table 5, and the formula glossaries disclose the matrix, scenarios, coefficients, optimizers, parameter-search budgets, sampling budget, and feasibility tests.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Table 2 reports 95% instance-level bootstrap intervals; Appendix C gives paired resampling, 10,000 resamples, and fixed inference seeds.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Appendix C records Python/NumPy versions, operating system, CPU allocation, memory, GPU absence, and a 27.64-second full-rerun wall time, while disclosing missing original timing metadata.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.

- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The study uses only synthetic data, avoids deployment and advantage overclaims, preserves double blindness, and discusses foreseeable allocation harms and mitigations in Appendix F.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Section 6 and Appendix F cover potential accountability benefits as well as automation, labor, fairness, privacy, and authority risks with mitigations.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: No model, personal dataset, high-risk dual-use asset, or deployable decision service is released; the synthetic benchmark is nevertheless bounded by explicit non-deployment guidance.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: The implementation snapshot is MIT licensed, the NeurIPS 2026 template is supplied by the venue, and scholarly methods are credited through the bibliography.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The synthetic generator, evaluator, frozen configuration, outputs, figure sources, reproduction script, limitations, and claim boundary are documented in the artifact and Appendices A–E.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The work uses seeded synthetic instances only and involves neither crowdsourcing nor human participants.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.

- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: No human-subjects research or identifiable personal data are involved, so IRB review is not applicable.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: LLMs are not an important, original, or non-standard component of the generator, evaluator, classical controls, quantum simulation, or statistical analysis.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.